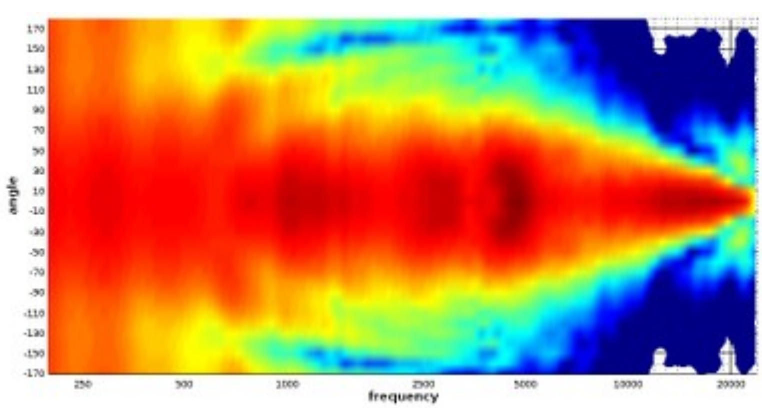
Constant Directivity behavior, without diffraction

The goal of a horn or waveguide is to control how a device—such as a tweeter or compression driver—radiates sound into the room.

This radiation pattern, or directivity, should remain consistent across frequencies.
However, when directivity is not constant, the coverage narrows progressively as the frequency increases. This can cause uneven sound dispersion and affect the listening experience.

To illustrate this, the polar plot below shows sound energy (in dB) as a function of the radiation angle:



On this polar map, which visualizes how sound energy spreads, you can see that the coverage becomes too narrow too early—starting around 5 kHz.
This narrowing makes the speaker sound “small” and incoherent because the sound radiates unevenly at different frequencies.

By addressing this issue—achieving constant directivity—we can

create a speaker with a coherent sound and a wide, natural soundstage.
This involves reducing effects like midrange narrowing and beaming, ensuring smooth directivity matching at the crossover, and maintaining proper center-to-center spacing according to crossover wavelength.

For detailed steps, you can follow our implementation guide.

Critical Distance, adapt your opening coverage to listening distance

We listen at a so-called critical distance where the direct field (sound coming directly from the speaker) and the reverberant field (sound that rebounds on walls) are approximately at a 50/50 or 60/40 ratio.

The brain has an integration time to merge these two fields, and this time varies according to frequency.

We choose a crossover point where directivity match occurs—that is, when the horn begins to lose its directivity and the woofer becomes more directive.
Thanks to this, there is no abrupt change in directivity.

Our main purpose is to **adapt opening coverage to listening distance** and room acoustics.

As seen in the critical distance article, it is important to avoid the "one-120°-coverage-fits-all" approach.

Here is a guide of coverage according to distance: